

Muhammad Afin Atsal

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Summary

AI/ML Engineer (Fresh Graduate, Information Technology GPA 3.64/4.00) with hands-on industry experience across two internships, specializing in Computer Vision and applied deep learning. Built and deployed end-to-end AI systems including a CNN-based visual inspection model (99.6% accuracy, 99.6% mAP@50) and RAG-based chatbots (FastAPI, PostgreSQL/PGVector, Google Gemini API) at PT XLSmart Telecom Sejahtera and PT Amman Mineral Internasional. Led thesis research on a two-stage YOLOv12 + HSCN pipeline for hierarchical waste detection, achieving 0.84 F1-score and 0.97 mAP. Proficient in Python, PyTorch, TensorFlow, OpenCV, and FastAPI, with proven ability to take models from prototype to production-integrated systems.

Education

University of Brawijaya

August 2022 – August 2026

Bachelor of Information Technology

- **GPA 3.64/4.00**

- **Thesis:** "Multi-Stage Waste Detection Model Based on YOLO and Hierarchical Multi-Label Classification" developed a two-stage pipeline (YOLOv12 + HSCN) achieving 0.84 F1-score (detection) and 0.97 mAP (classification)
- **Related Coursework:** Machine Learning, Neural Networks, Computer Vision, Large Language Models (LLMs), Artificial Intelligence, Data Mining, Big Data Analytics, Data Structures & Algorithms, Data Visualization, Database Systems, Database Programming, Software Engineering, DevOps Engineering

Professional Experience

X-Camp, PT XLSmart Telecom Sejahtera Tbk

Aug 2025 – Dec 2025

AI/ML Engineer (MBKM Program)

Jakarta, Indonesia

- Developed 3 AI/IoT solutions under X-Camp's Open Lab (AI & IoT), following Agile-Scrum methodology with weekly sprints.
- Built a computer vision-based Visual Inspection system using CNN to automate defect detection on production lines, achieving 99.6% accuracy, 95.8% precision, 100% recall, and 99.6% mAP@50; integrated with a real-time ThingsBoard dashboard for production monitoring.
- Designed and implemented an Object Removal Detection system using OpenCV template matching (multi-scale, normalized cross-correlation) to monitor critical safety equipment, with state-detection logic (Detected/Partially Blocked/Missing) and real-time alerting.
- Developed a RAG-based internal chatbot using FastAPI, BAAI/bge-m3 embeddings, and Gemini Flash 2.5 to enable document-based Q&A over company data, improving information access efficiency.

IT Department, PT Amman Mineral Internasional Tbk

Jul 2025 – Aug 2025

Back-End & AI System Developer (Internship)

Sumbawa Barat, Indonesia

- Built the back-end and AI system for an interactive RAG-based chatbot to automate boat ticket booking within the company's internal SuperApps ecosystem.
- Designed a modular microservices architecture using FastAPI, integrating real-time scheduling data with PostgreSQL + PGVector for semantic search and retrieval.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with Google Gemini API, supporting bilingual (Indonesian/English) multi-turn conversations with contextual memory.
- Achieved average API latency below 2 seconds post-initial embedding; validated system through Postman API testing and Flutter front-end integration.

Teaching Experience

Fakultas Ilmu Komputer (FILKOM) & Fakultas MIPA, Universitas Brawijaya

Feb 2026 – Jun 2026

Python Practicum Instructor (Joint Program)

Malang, Indonesia

- Selected as one of the practicum instructors in a cross-faculty collaboration between FILKOM and FMIPA, teaching Python programming to students for one semester.
- Co-developed curriculum and practicum materials alongside instructors from both departments.
- Evaluated student assignments and provided technical mentoring throughout the semester.

Freelance

Private Tutor – IoT & Applied AI Systems

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Malang, Indonesia

- Mentored high school students in developing an E-Nose (Electronic Nose) system for their scientific research competition (Karya Tulis Ilmiah), detecting volatile compounds (ethanol/alcohol) in cattle breath to identify early bacterial infection indicators.
- Taught Python programming and embedded systems development (ESP32, Arduino), including sensor integration and circuit assembly.
- Guided the full pipeline: sensor data acquisition, preprocessing, and AI-based classification for health indication.

Organizational and Event Experience

KBMSI FILKOM UB

Deputy Head of Entrepreneurship Department

Feb 2024 – Dec 2024

Malang, Indonesia

- Successfully executed 4 main programs, including a seminar and business plan competition aimed at fostering student entrepreneurship.
- Initiated and managed the DSI Store program, producing and selling merchandise (e.g., t-shirts, stickers), generating profit to support organizational funding.
- Led the design and production of official PDH (Organizational Jacket) for KBMSI members, enhancing organizational identity and unity.
- Created and managed content for entrepreneurial-themed social media posts, supporting brand visibility and knowledge sharing.
- Actively contributed to strategic planning to improve department performance and organizational sustainability.

KBMSI FILKOM UB

Staff of Entrepreneurship Department

Feb 2024 – Dec 2024

Malang, Indonesia

- Contributed to the DSI Store merchandise production and sales, supporting organizational revenue generation.
- Assisted in planning and executing entrepreneurial events and contributed to internal coordination for smooth program delivery.

FILAFEST (Filkom Anniversary Festival)

Head of Sponsorship and Fundraising

Apr 2023 – Dec 2023

Malang, Indonesia

- Developed and led sponsorship acquisition strategies, successfully securing partnerships with multiple external stakeholders.
- Managed end-to-end proposal development, negotiation, and reporting, ensuring professional communication and mutual value delivery with sponsors.
- Collaborated with other divisions to allocate funds efficiently across events and support the overall success of FILAFEST 2023.
- Raised funds exceeding initial targets, contributing significantly to the festival's operational budget.

Project & Achievements

Multi-Stage Waste Detection using YOLOv12 and HSCN

AI Engineering & Computer Vision (Undergraduate Thesis)

Feb 2026 – Jul 2026

Malang, Indonesia

- Designed a two-stage pipeline combining YOLOv12 for object detection and a Hierarchical Sequential Classification Network (HSCN) for three-level hierarchical waste classification (management status, material type, specific object).
- Aggregated and curated a custom dataset from TrashNet, Kaggle, TACO, and RealWaste (2,116 images, 2,582 bounding box annotations).
- Benchmarked 3 YOLOv12 variants (S/M/L) and 6 HSCN backbones (ResNet18/50/101, EfficientNet-B3, ConvNeXt-Small, MobileNetV3-Large); best results: YOLOv12l (F1 = 0.8440, mAP@0.5:0.95 = 0.7636) and ConvNeXt-Small (mAP_mean = 0.9670).
- Built a Flask-based inference application with a backbone comparison mode for real-time model evaluation and thesis defense demonstration.

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AI Engineering

Feb 2024 – May 2025

Malang, Indonesia

- Developed an AI grammar checking model using a fine-tuning approach with LLaMA 3, aligned with target outputs.
- Conducted preprocessing, tokenization, and training using domain-specific dataset (Grammarly CoEdit).
- Evaluated model performance using BLEU and edit distance metrics.

Pneumonia Detection from Chest X-Ray Images Using Convolutional Neural Networks

Mar 2024

AI Engineering & Computer Vision (Personal Project)

Malang, Indonesia

- Developed a convolutional neural network (CNN) to classify chest X-ray images for pneumonia detection with high sensitivity and specificity.
- Conducted data preprocessing including normalization, data augmentation, and class balancing to handle imbalanced datasets.
- Designed and trained a custom CNN architecture with dropout and batch normalization to prevent overfitting.
- Achieved over 92% validation accuracy on the test dataset, outperforming baseline models.
- Visualized learning curves and confusion matrices to interpret model performance and support clinical validation.

Fraud Detection Big Data

Oct 2024

Data Engineer (Personal Project)

Malang, Indonesia

- Developed a fraud detection pipeline using Python and Pandas on large-scale transaction, user, and card datasets.
- Performed full ETL process: cleaned missing values, encoded categorical features, and normalized data for analysis.
- Engineered features from transaction metadata (amounts, timestamps, device usage) to enhance fraud pattern visibility.
- Applied unsupervised learning (K-Means) to detect anomalous transaction clusters as potential fraud.
- Prepared data for classification modeling (fraud vs. non-fraud) using label data and stratification indicators.

Cognitive Performance Prediction based on Wearable Exam Stress Data

Feb 2025 – Jun 2025

AI Engineering & Data Analytics

Malang, Indonesia

- Developed a machine learning pipeline for predicting students' exam scores based on biosignal data (HR, EDA, TEMP, ACC) collected from wearable sensors (Empatica E4).
- Implemented complete data preprocessing including signal denoising, sliding window segmentation, feature extraction, and normalization.
- Built and tuned a Random Forest Regressor to map physiological features to academic performance, achieving R^2 of 0.68 on the test set.

Certifications

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| • Building AI Agents and Agentic Workflows Specialization - IBM | Aug 2026 |
| • Machine Learning Specialization - DeepLearning.AI & Stanford Online | Aug 2026 |
| • Google AI Professional Certificate - Google | Aug 2026 |
| • Google AI Essentials Specialization - Google | Aug 2026 |

Skills

- **Hard skills:** Machine Learning (Advanced), Computer Vision (Advanced), Deep Learning (Advanced), Natural Language Processing & LLMs (Intermediate-Advanced), Retrieval-Augmented Generation (RAG) (Advanced), Data Analysis (Intermediate-Advanced), Backend Development (Intermediate), ETL & Data Preprocessing (Intermediate), Data Visualization (Advanced), Statistical Testing.
- **Soft skills:** Leadership, Critical Thinking, Problem-Solving, Adaptability, Teamwork and Collaboration, Time Management, Project Coordination, Analytical Thinking, Communication & Technical Mentoring.
- **Tools and Tech stack:** Python, PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, YOLO, LangChain, FastAPI, Gradio, PostgreSQL, PGVector, Docker, Flutter, Firebase, Git, GitHub, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, SQL, Apache Spark, Hadoop, ThingsBoard, VS Code, Jupyter Notebook, Google Colab.